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CAVEAT: Reliability-aware active verification under correlated errors for multi-agent LLM reasoning

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Abstract

Multi-agent large language model (LLM) systems improve reasoning reliability by letting several agents solve a task and reconciling their answers through debate or voting, often followed by a verification stage that checks intermediate steps with additional judges or tools. These pipelines assume that agents and verifiers err independently. In practice, agents from the same model family share training data and failure modes, so their verdicts are correlated and repeated same-family checks carry far less evidence than independence suggests. This paper treats verification as a resource-allocation problem under correlated errors and proposes CAVEAT (Correlation-Aware VErification AllocaTion). CAVEAT merges heterogeneous agent traces into a claim dependency graph, models each verifier by Beta-distributed sensitivity and specificity together with a family-level shared-slip rate that yields a closed-form correlated posterior, and greedily selects which claim to verify, by which verifier and when to stop by a cost-normalised value of information. Ignoring correlation inflates posterior log-odds by a design-effect factor $1+(k-1)\rho$, and the greedy policy is near-optimal under adaptive submodularity. On six public benchmarks covering claim verification, multi-hop question answering and mathematical reasoning, with eight heterogeneous verifiers, CAVEAT matches exhaustive verification at 21% of its cost, improves accuracy over the strongest budget-matched baseline by 3.1 points, reduces expected calibration error from 7.4% to 3.8% and lowers the rate of confidently accepted wrong answers from 19.6% to 7.2%.

Keywords: Multi-agent systems, Large language models, Active verification, Correlated errors, Value of information, Reliability estimation

1 Introduction

Large language model (LLM) agents have moved from single-turn assistants to systems that decompose a task, call tools, and coordinate several roles to produce an answer (Hong et al. 2024; Wang et al. 2024; Wu et al. 2023; Yao et al. 2023). A

widely adopted recipe for making such systems more reliable is redundancy: multiple agents solve the same problem, and their outputs are reconciled by majority voting (Wang et al. 2023), multi-agent debate (Du et al. 2024; Khan et al. 2024; Liang et al. 2024), or an LLM judge (Chan et al. 2024; Zheng et al. 2023). Redundancy is then frequently complemented by an explicit verification stage: intermediate claims are checked against retrieved evidence (Asai et al. 2024; Lewis et al. 2020; Min et al. 2023), re-derived by a second model (Cohen et al. 2023; Dhuliawala et al. 2024; Weng et al. 2023), or executed by a program (Chern et al. 2023; Gou et al. 2024). Verification has become the standard answer to the hallucination problem in agentic settings (Huang et al. 2025; Matharaarachchi et al. 2026), and debate-and-adjudication or coordinator-checker layouts have recently been adopted for misinformation detection (Luo et al. 2026), question generation (Saadaoui and Alonso 2025), structured QA (Lee and Lee 2025) and governance of agent decisions (Antuley et al. 2026).

Redundancy and verification both pay off only if the redundant components fail in different ways. This condition is rarely examined. Two observations motivate the present work. First, the benefit of debate over plain voting is smaller than commonly assumed: Choi et al. (2025) show, across seven benchmarks, that majority voting accounts for most of the gain attributed to debate, and prove that unbiased debate induces a martingale over agent beliefs and hence cannot improve expected correctness. Second, the agents and verifiers used in practice are not independent samples. They are drawn from a handful of model families that share pre-training corpora, instruction-tuning recipes and, consequently, systematic misconceptions. When the solving agents share a wrong intermediate fact, the "consensus" that debate or voting produces is a confabulation consensus, and a verifier from the same family is likely to confirm it. A pipeline that treats such verdicts as independent evidence becomes confidently wrong, and downstream components or human operators act on that confidence.

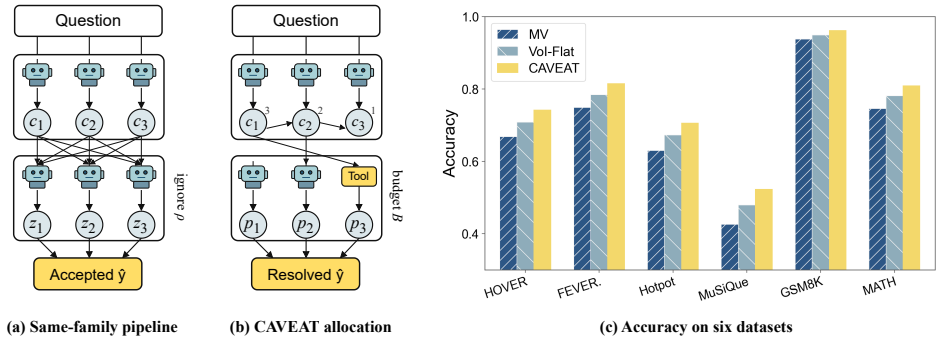


Fig. 1 Motivation of correlation-aware active verification. (a) A same-family pipeline aggregates the verdicts of densely connected judges as if their error correlation ρ were zero. (b) CAVEAT organises the claims as a dependency graph (superscripts denote depth) and spends a budget B on heterogeneous judges and an external tool. (c) Accuracy of majority voting (MV), dependency-agnostic value-of-information allocation (VoI-Flat) and CAVEAT on six datasets under the same solvers and budget

Fig. 1 summarises the failure mode and its empirical consequence. In the same-family pipeline of panel (a), several agents produce candidate claims and several judges assess them, but repeated verdicts are treated as independent even when they share the same error source. Panel (b) replaces this dense, indiscriminate checking pattern with CAVEAT: the claim graph exposes which claims support later steps, and the budget is routed to a complementary verifier or tool where it has the largest expected effect. Panel (c) gives the resulting accuracy on six datasets. Under the same verification budget, CAVEAT improves over majority voting by 2.4–9.7 percentage points and over dependency-agnostic VoI-Flat by 1.3–4.4 points; the largest gain occurs on MuSiQue, whose longer reasoning chains make misplaced verification particularly costly.

The question is therefore how to spend a bounded verification budget when the available verifiers are unequally reliable, unequally priced and correlated. We frame this as an active verification problem with three coupled decisions: which reasoning claim to check, by which verifier, and when to stop. Prior work addresses these decisions in isolation. Process reward models (Lightman et al. 2024) and self-verification prompts (Ling et al. 2023; Weng et al. 2023) check every step with a single verifier;

confidence-gated schemes verify the least confident claim (Xiong et al. 2024); value-of-information (VoI) selection is used for anytime verification but assumes a single trusted checker (Howard 1966; Krause and Guestrin 2009). None of them model the reliability of the verifier, and none account for correlation between verifiers or between verifiers and the agents that produced the claim.

To this end, we propose CAVEAT (Correlation-Aware VErification AllocaTion). The reasoning traces of heterogeneous solver agents are decomposed into atomic claims and merged into a dependency graph, in which each claim receives a *downstream mass* that counts how much of the final answer rests on it, so that errors near the root of the derivation are prioritised over errors in leaves. Every verifier is described by a sensitivity and a specificity with Beta posteriors estimated from a small calibration set, and every model family by a *shared-slip rate*, the probability that all members of the family make the same error on a claim. The resulting mixture likelihood has a closed form and reduces to the Dawid–Skene model (Dawid and Skene 1979) when the slip rate is zero and to a single verifier when it is one; the same construction applied to the solver agents gives a correlation-corrected prior for each claim. A greedy allocation loop then scores every (claim, verifier) pair by the expected reduction of dependency-weighted uncertainty per unit cost, queries the best pair, updates the correlated posterior, and stops when the residual uncertainty of the final answer falls below a threshold or the budget is exhausted; if the budget runs out while the answer is still uncertain, the system abstains and escalates the instance.

The contributions of this paper are as follows.

- We formalise the verification of multi-agent reasoning as budgeted allocation over (claim, verifier) pairs under correlated errors and show that the independence assumption inflates posterior log-odds by a factor $1 + (k - 1)\rho$ for k same-family verdicts with within-family correlation ρ.

- We propose a family-shared-slip reliability model with a closed-form posterior, a dependency-weighted risk functional over the merged claim graph, and a cost-normalised greedy allocation policy with a $(1-1/e)$ approximation guarantee under adaptive submodularity. The method does not depend on the solver framework and can wrap ReAct, debate or supervisor-style systems.
- We evaluate CAVEAT on six public benchmarks in three task families with four heterogeneous solver agents and eight verifiers. It reaches the accuracy of exhaustive verification at about one fifth of the cost, improves on the strongest budget-matched baseline by 3.1 accuracy points on average, halves the expected calibration error and reduces false certainty by a factor of 2.7. We also measure the error correlation present in the pool and analyse allocation and stopping behaviour, robustness, cross-backbone transfer, hyper-parameter sensitivity and inference cost.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 formalises the problem and analyses the cost of ignoring correlation. Section 4 presents CAVEAT. Section 5 reports the experiments, Section 6 discusses limitations, and Section 7 concludes.

2 Related work

2.1 Multi-agent reasoning, debate and voting

Chain-of-thought prompting (Wei et al. 2022) made intermediate reasoning visible and self-consistency (Wang et al. 2023) exploited it by sampling several chains and voting on the answer. Du et al. (2024) extended sampling to communication: agents read each other's answers and revise, which improved factuality and arithmetic accuracy. Follow-ups vary the protocol (divergent-thinking judges (Liang et al. 2024), persuasion asymmetry (Khan et al. 2024), evaluator debates (Chan et al. 2024)) and the orchestration substrate (Hong et al. 2024; Wu et al. 2023); surveys of the area are given in

(Guo et al. 2024; Wang et al. 2024). Two recent lines of analysis are directly relevant to us. Estornell and Liu (2024) formalise debate and show that shared biases can be amplified rather than cancelled; Choi et al. (2025) disentangle voting from debate and find the latter contributes little. Chen et al. (2024) show that more calls to the same model do not monotonically help. Failure taxonomies of multi-agent systems (Cemri et al. 2025) list "incorrect verification" and "premature termination" among the dominant failure categories, and failure-attribution benchmarks (Zhang et al. 2025) report that even strong models locate the decisive wrong step only 14% of the time. These negative results are the starting point of our work: since debate does not correct shared errors, external verification has to, and its allocation should account for the correlation structure of the pool.

2.2 Verification and self-correction of LLM reasoning

Verification of LLM outputs comes in three forms. Model-based verification uses a second LLM as a judge (Zheng et al. 2023), a cross-examiner (Cohen et al. 2023), or a process reward model trained on step-level labels (Lightman et al. 2024); self-verification (Ling et al. 2023; Weng et al. 2023), chain-of-verification (Dhuliawala et al. 2024) and feedback loops such as Self-Refine (Madaan et al. 2023) and Reflexion (Shinn et al. 2023) let the same model check itself, with limited success on hard reasoning (Huang et al. 2024; Stechly et al. 2024). LLMs do carry some signal about what they know (Kadavath et al. 2022), but verbalised confidence is over-confident on exactly the instances where several models share a wrong belief (Xiong et al. 2024). Evidence-based verification grounds atomic claims in retrieved documents (Asai et al. 2024; Lewis et al. 2020; Min et al. 2023). Tool-based verification executes code or calls calculators (Chern et al. 2023; Gou et al. 2024). These verifiers differ considerably in cost and coverage; a program checker, for instance, is near-perfect on arithmetic but silent on encyclopaedic facts. Existing pipelines either verify every step with one

verifier or verify the least confident step (Tian et al. 2023; Xiong et al. 2024). Test-time compute scaling (Snell et al. 2025) studies how to split a budget between more samples and more verification but keeps a single verifier. Our contribution is complementary: given a heterogeneous pool of imperfect verifiers, we decide how to route claims to them.

2.3 Truth inference from unreliable sources and value of information

Estimating item labels from multiple noisy annotators is a classical problem. Dawid and Skene (1979) model each annotator by a confusion matrix and fit it by EM; Raykar et al. (2010) and Whitehill et al. (2009) add learning and item difficulty. These models assume annotators are conditionally independent given the true label. Kuncheva et al. (2003) derive the limits of majority voting for dependent classifiers, and the design effect of Kish (1965) quantifies how correlated samples reduce effective sample size. Deciding which observation to acquire next is the value-of-information problem (Howard 1966; Krause and Guestrin 2009), whose greedy solution enjoys approximation guarantees when the objective is (adaptive) submodular (Golovin and Krause 2011; Nemhauser et al. 1978), a property widely used in active learning (Settles 2012). CAVEAT brings these tools to the multi-agent setting: verifiers are annotators with learned reliability, model families induce correlation, and the graph structure of the reasoning trace determines the utility of resolving each claim.

2.4 Position of this work

Table 1 contrasts CAVEAT with representative systems along the decisions that a verification layer must make. Debate-based misinformation detectors such as JusticeFact (Luo et al. 2026) and D2D (Han et al. 2025) verify by adversarial exchange among same-backbone roles; the dual-memory knowledge-graph agent of Matharaarachchi

Table 1 Comparison of reasoning-verification systems

Method	Multi-ag.	Claim sel.	Verif. sel.	Reliab.	Corr.	Budget	Signal
Self-consistency (Wang et al. 2023)	sampling	–	–	–	–	–	answer vote
MAD (Du et al. 2024)	✓	–	–	–	–	–	peer answers
Self-verif. (Weng et al. 2023)	–	all steps	–	–	–	–	same LLM
PRM (Lightman et al. 2024)	–	all steps	–	–	–	–	trained scorer
CoVe (Dhuliawala et al. 2024)	–	planned	–	–	–	–	same LLM
CRITIC (Gou et al. 2024)	–	all	fixed tool	–	–	–	tool feedback
JusticeFact (Luo et al. 2026)	✓	–	–	partial	–	–	debate+ web
Dual-memory KG (Matharaarachchi et al. 2026)	–	trace	cascade	–	–	–	KG+ human
SORA-ATMAS (Antuley et al. 2026)	✓	–	–	✓	–	–	multi-LLM MAE
CAVEAT (ours)	✓	✓	✓	✓	✓	✓	8 heterog. verifiers

Notes: Multi-ag.: multiple solvers; Claim sel.: claim selection; Verif. sel.: verifier selection; Reliab.: reliability estimation; Corr.: error correlation; Budget: budgeted verification with stopping; Signal: verification source.

et al. (2026) validates traces through a fixed cascade; SORA-ATMAS (Antuley et al. 2026) gates the outputs of multiple LLMs through trust thresholds but treats them as independent; the engineering-modelling agents of Liu et al. (2026) rely on a fixed human-in-the-loop review stage. To our knowledge, no existing method jointly selects the claim, the verifier and the stopping time while modelling verifier reliability and inter-verifier correlation.

Table 2 Main notation

Symbol	Meaning
$\mathcal{A}$, a, $F(a)$	solver agents, one agent, its model family
$G = (\mathcal{C}, E)$, c_i, y_i	merged claim graph, claim i, its latent truth
$\hat{y}$	answer node; candidate answers $a \in \mathcal{Y}$
w_i	downstream mass (propagation weight) of claim c_i
$\mathcal{V}$, v_j, $F(j)$, c_j	verifier pool, one verifier, its family, its unit cost
α_j, β_j	sensitivity $P(z = 1 \mid y = 1)$ and specificity $P(z = 0 \mid y = 0)$ of v_j
ρ_F, α_F, β_F	shared-slip rate of family F and its family-level sensitivity and specificity
z_{ij}, Z_i	verdict of v_j on c_i; all verdicts collected on c_i
$p_i = P(y_i = 1 \mid Z_i)$	current posterior of claim c_i
$H_\mathrm{b}(p)$	binary entropy $-p \log p - (1-p)\log(1-p)$
$\mathcal{R}$	dependency-weighted risk $\sum_i w_i H_\mathrm{b}(p_i)$
Δ_{ij}	expected risk reduction from querying (c_i, v_j)
B, ϵ, τ	budget, stopping threshold, abstention threshold
κ_0, λ	Beta prior strength, propagation exponent

3 Problem formulation

3.1 Setting and notation

A task x is given to a set of solver agents $\mathcal{A} = \{a_1, \ldots, a_n\}$, each producing a reasoning trace and a final answer. Traces are decomposed into atomic claims and merged into a claim graph $G = (\mathcal{C}, E)$ with $\mathcal{C} = \{c_1, \ldots, c_N\}$ and an edge $(c_k \to c_i) \in E$ when claim c_i is derived from claim c_k. A distinguished answer node $\hat{y}$ has as parents the claims that directly support each candidate answer. Every claim has a latent truth value $y_i \in \{0, 1\}$. A pool of verifiers $\mathcal{V} = \{v_1, \ldots, v_m\}$ is available; querying v_j on c_i costs c_j and returns a verdict $z_{ij} \in \{0, 1, \varnothing\}$, where $\varnothing$ denotes abstention (e.g. a program checker asked about a historical fact). Verifiers are partitioned into families $F(j)$ (e.g. OpenAI, Qwen, Llama, retrieval, program), and solver agents into families $F(a)$. Given a budget B, the system chooses a sequence of queries and finally commits to an answer or abstains. Table 2 lists the symbols.

3.2 The cost of assuming independence

Consider a single claim with prior odds O_0 and k verdicts from verifiers of the same family, each with accuracy $a > 1/2$ and symmetric error ($\alpha_j = \beta_j = a$). Under conditional independence the posterior log-odds after k agreeing verdicts is $\log O_0 + k\,\ell$ with $\ell = \log \frac{a}{1-a}$: evidence grows linearly in k. Now let the errors be exchangeably correlated with pairwise correlation ρ, which is the failure mode depicted in Fig. 1(a). A standard design-effect argument (Kish 1965) gives the following.

Proposition 1 (False certainty under ignored correlation) *Let k verdicts have common accuracy a and exchangeable pairwise error correlation $\rho \in [0, 1]$. The information carried by the k verdicts equals that of $k_{\text{eff}} = k/(1 + (k-1)\rho)$ independent verdicts, so the independence model overstates the posterior log-odds increment by the factor $1 + (k-1)\rho$. In particular, as $k \to \infty$ the true posterior log-odds saturates at $\log O_0 + \ell/\rho$, whereas the independence model diverges.*

Proof sketch Write the k error indicators as $e_j = \mu + \eta_j$ with $\mathrm{Var}(\bar{e}) = \frac{\sigma^2}{k}(1 + (k-1)\rho)$. The mean error rate estimated from k correlated indicators has the variance of k_{eff} independent ones; the log-likelihood ratio contributed by the verdicts is proportional to the precision of that estimate, giving the stated factor. The saturation follows from $k_{\text{eff}} \to 1/\rho$. □

With $\rho = 0.3$, roughly the within-family pairwise error correlation measured on our calibration claims, and $k = 3$, the independence model reports 1.6× the evidence actually collected. The same argument applies to solver agents: three same-family agents agreeing on a wrong intermediate fact do not amount to three votes, and a verification layer that starts from a vote-based prior inherits the inflation unless it is corrected. CAVEAT therefore applies the correlation model to the solver votes as well.

4 The CAVEAT framework

4.1 Overview

Fig. 2 depicts the four steps of CAVEAT on a three-hop HOVER instance. Step 1 runs the heterogeneous solver agents once. Step 2 extracts atomic claims, merges duplicates across agents, builds dependency edges and computes a downstream mass for every claim; the correlation-corrected solver votes give each claim a prior. Step 3 holds the verifier pool with its reliability model, calibrated offline once per task family. Step 4 is the active loop: score all admissible (claim, verifier) pairs by cost-normalised value of information, query the best one, update the correlated posterior, and repeat until the answer posterior is sharp enough or the budget is spent. The loop only consumes claims, edges and votes, so it can be attached to ReAct agents, debate protocols or supervisor pipelines without modifying them.

4.2 Claim graph construction

Each solver trace is passed to a claim extractor (GPT-4o-mini, prompt in Appendix A) that returns a list of atomic, self-contained statements together with, for each statement, the indices of the earlier statements it was derived from. This is the same decomposition used by FActScore (Min et al. 2023) for factual precision, with the addition of derivation pointers. Claims from different agents are merged when the cosine similarity of their bge-large embeddings (Xiao et al. 2024) exceeds 0.85 and an NLI model (He et al. 2023) labels the pair as bidirectionally entailed; contradictory pairs (e.g. "born in Illinois" vs "born in California") are kept as separate nodes linked by a conflict edge so that at most one of them can be true. Edges from all traces are unioned. The answer node $\hat{y}$ collects, for every distinct candidate answer a, the set $\mathrm{Pa}(a)$ of claims each agent used as the last step before stating a. The resulting graph typically has 4–9 claims (mean 6.3 over our datasets) and is a DAG by construction. Fig. 3 shows the three stages on one solver trace of the running example of Fig. 2: the

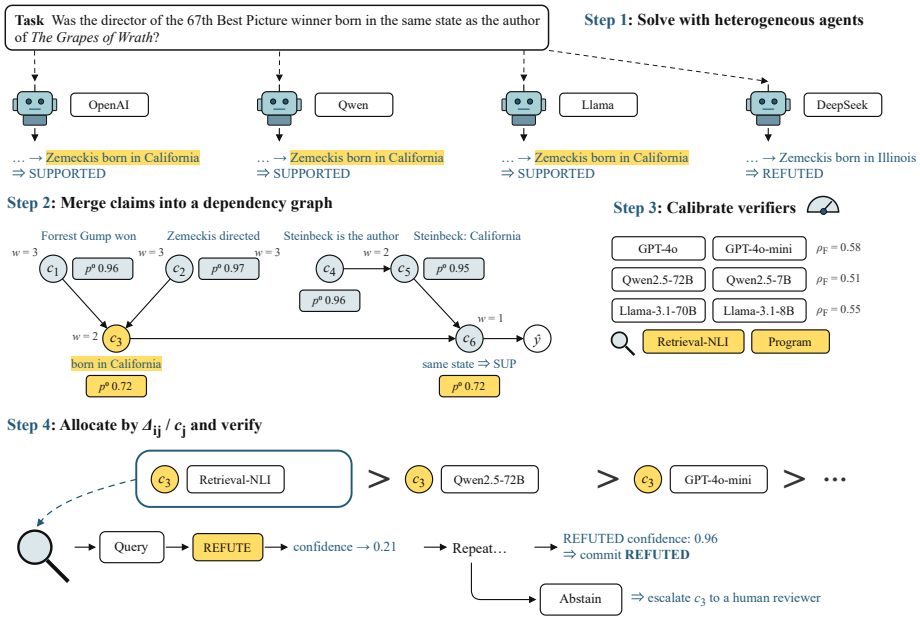


Fig. 2 Overview of CAVEAT on a three-hop HOVER instance: heterogeneous solving (Step 1), claim graph construction with downstream masses w_i and priors $p_i^{(0)}$ (Step 2), the calibrated verifier pool with shared-slip rates ρ_F (Step 3), and the allocation loop over (claim, verifier) pairs scored by Δ_{ij}/c_j (Step 4)

trace, the extractor's JSON output with derivation pointers, and the merged graph with its masses and priors.

4.2.1 Downstream mass

An error in a claim that many later claims rely on is worse than an error in a leaf. We quantify this with the downstream mass

$$w_i = \Big(1 + \sum_{k:\,(c_i \to c_k) \in E} w_k\Big)^{\lambda}, \qquad w_{\hat{y}} = 0, \tag{1}$$

computed by a reverse topological pass. With $\lambda = 1$, w_i counts the number of derivation paths from c_i that reach the answer; $\lambda < 1$ compresses the range and $\lambda = 0$ recovers a flat weighting. Step 2 of Fig. 2 shows the masses for a three-hop instance.

(a) Solver trace of a_2 (Qwen2.5-72B)

TRACE The Best Picture winner at the 67th Academy Awards (1995) was *Forrest Gump*. *Forrest Gump* was directed by Robert Zemeckis. Zemeckis is a Hollywood director who was born in California. *The Grapes of Wrath* was written by John Steinbeck. Steinbeck was born in Salinas, California. Both men were born in California, so the claim holds. `Final answer: SUPPORTED`

a_1 (GPT-4o-mini), a_3 (Llama-3.1-70B): same highlighted fact, SUPPORTED. a_4 (DeepSeek-V3): "born in Chicago, Illinois", REFUTED.

claim extraction (GPT-4o-mini, temperature 0)

(b) Extracted claims (JSON)

```
[{"id":1, "type":"retrieval_fact", "derived_from":[],
   "claim":"Forrest Gump won Best Picture at the 67th Academy Awards."},
{"id":2, "type":"retrieval_fact", "derived_from":[],
   "claim":"Robert Zemeckis directed Forrest Gump."},
{"id":3, "type":"retrieval_fact", "derived_from":[1,2],
   "claim": "Robert Zemeckis was born in California." },
{"id":4, "type":"retrieval_fact", "derived_from":[],
   "claim":"John Steinbeck wrote The Grapes of Wrath."},
{"id":5, "type":"retrieval_fact", "derived_from":[4],
   "claim":"John Steinbeck was born in Salinas, California."},
{"id":6, "type":"logical", "derived_from":[3,5],
   "claim":"The director and the author were born in the same state."}]
```

merge across the four agents; compute w_i, $p_i^{(0)}$

(c) Merged claim graph

	Claim (merged)	Asserted by	w_i	$p_i^{(0)}$
c_1	Forrest Gump won Best Picture at the 67th Academy Awards	all four	3	0.96
c_2	Robert Zemeckis directed Forrest Gump	all four	3	0.97
c_3	Robert Zemeckis was born in California	a_1, a_2, a_3	2	0.72
c_4	John Steinbeck wrote The Grapes of Wrath	all four	3	0.96
c_5	John Steinbeck was born in Salinas, California	all four	2	0.95
c_6	Director and author born in the same state $\Rightarrow$ SUPPORTED	a_1, a_2, a_3	1	0.72

Not listed: the conflicting node of a_4 ("born in Illinois"), linked to c_3 by a conflict edge, $p^{(0)} = 0.28$.

Fig. 3 Claim graph construction on the running HOVER example. (a) Trace of the Qwen2.5-72B solver, with the wrong intermediate fact highlighted. (b) Output of the claim-extraction prompt (Prompt A.1). (c) Merged claim graph with asserting agents, downstream masses w_i and correlation-corrected priors $p_i^{(0)}$

4.2.2 Correlation-corrected prior

Let $s_{ia} \in \{0, 1, \varnothing\}$ indicate whether agent a asserted c_i, asserted a conflicting claim, or was silent. Treating the agents as noisy annotators with family-shared slips (the same model as for verifiers), the prior $p_i^{(0)} = P(y_i = 1 \mid \{s_{ia}\})$ follows from Eq. (3) with agent parameters (α_a, β_a) and agent-family parameters $(\rho_F, \alpha_F, \beta_F)$ estimated on the calibration set. Claims asserted by all four agents start with $p^{(0)} \approx 0.95$. A claim asserted by three agents and contradicted by the fourth starts lower (0.72 in the example, against the 0.75 of a plain vote share), because the slip mass of each asserting family is not counted as independent evidence and the dissenting agent, DeepSeek-V3, is the one least coupled to the others.

4.3 Correlated reliability model

Each verifier v_j has a sensitivity $\alpha_j = P(z_{ij} = 1 \mid y_i = 1)$ and a specificity $\beta_j = P(z_{ij} = 0 \mid y_i = 0)$ with Beta posteriors $\alpha_j \sim \text{Beta}(u_j, u'_j)$, $\beta_j \sim \text{Beta}(r_j, r'_j)$ initialised from a prior of strength κ_0 centred at 0.75 and updated by counting hits and misses on the calibration set. Coverage is handled by an abstention probability per (verifier, claim type) pair; abstentions carry no evidence and cost the verifier's price.

Conditional independence across verifiers of the same family is replaced by a shared-slip mixture. For family F with member verdicts $Z_{iF} = \{z_{ij} : F(j) = F\}$ on claim c_i, we introduce a latent indicator $s_{iF} \sim \text{Bernoulli}(\rho_F)$. If $s_{iF} = 1$ the whole family behaves as a single verifier: all members emit the same verdict $\bar{z}$, drawn from a family-level confusion with sensitivity α_F and specificity β_F. If $s_{iF} = 0$ the members behave independently with their own (α_j, β_j). The family likelihood is therefore

$$\begin{aligned} P(Z_{iF} \mid y_i) = \rho_F \, \mathbb{I}[\text{all } z_{ij} \in Z_{iF} \text{ equal}] \, P(\bar{z} \mid y_i; \alpha_F, \beta_F) \\ + (1 - \rho_F) \prod_{j \in F} P(z_{ij} \mid y_i; \alpha_j, \beta_j), \end{aligned} \tag{2}$$

Algorithm 1 Correlated posterior update for claim c_i

Require: prior $p_i^{(0)}$; collected verdicts Z_i grouped by family; parameters $\{\alpha_j, \beta_j\}$ (posterior means), $\{\rho_F, \alpha_F, \beta_F\}$

for $y \in \{0, 1\}$ **do**
  $L_y \leftarrow 1$
  for each family F with non-abstaining verdicts $Z_{iF} \neq \emptyset$ **do**
    $L^{\text{ind}} \leftarrow \prod_{j \in F} \left[y\,(\alpha_j^{z_{ij}}(1-\alpha_j)^{1-z_{ij}}) + (1-y)\,(\beta_j^{1-z_{ij}}(1-\beta_j)^{z_{ij}}) \right]$
    $L^{\text{slip}} \leftarrow \mathbb{I}[\text{all } z_{ij} \text{ equal}] \left[y\,(\alpha_F^{\bar{z}}(1-\alpha_F)^{1-\bar{z}}) + (1-y)\,(\beta_F^{1-\bar{z}}(1-\beta_F)^{\bar{z}}) \right]$
    $L_y \leftarrow L_y \cdot \left(\rho_F L^{\text{slip}} + (1-\rho_F) L^{\text{ind}}\right)$
  end for
end for
$p_i \leftarrow p_i^{(0)} L_1 / \left(p_i^{(0)} L_1 + (1 - p_i^{(0)}) L_0\right)$
renormalise p_i against claims linked to c_i by conflict edges
return p_i

where $P(z \mid y; \alpha, \beta)$ is the usual confusion term, $\alpha^z(1-\alpha)^{1-z}$ for $y = 1$ and $\beta^{1-z}(1-\beta)^z$ for $y = 0$. Families are conditionally independent of each other given y_i, so

$$p_i = P(y_i = 1 \mid Z_i) = \frac{p_i^{(0)} \prod_F P(Z_{iF} \mid y_i = 1)}{p_i^{(0)} \prod_F P(Z_{iF} \mid y_i = 1) + (1 - p_i^{(0)}) \prod_F P(Z_{iF} \mid y_i = 0)}. \quad (3)$$

The model has a closed form, so no sampling is needed and the update in Algorithm 1 costs $O(m)$ per claim. It interpolates between known models, since $\rho_F = 0$ recovers Dawid–Skene (Dawid and Skene 1979) and $\rho_F = 1$ collapses the family to one verifier. Note that ρ_F is the probability that the family moves together, not the correlation of its errors: the common verdict is usually correct, so the pairwise error correlation induced by the mixture is well below ρ_F (about 0.3–0.4 for the families used here). ρ_F is initialised by matching the observed agreement rate of the family and refined by maximum likelihood together with (α_F, β_F) (Appendix B). We use $\rho_{\text{OpenAI}} = 0.58$, $\rho_{\text{Qwen}} = 0.51$, $\rho_{\text{Llama}} = 0.55$ and treat the retrieval and program checkers as singleton families. Conflict edges are enforced by renormalising the posteriors of mutually exclusive claims to sum to at most one after every update.

4.4 Dependency-weighted risk

The state of the system is summarised by the risk functional

$$\mathcal{R}(\{p_i\}) = \sum_{i=1}^{N} w_i \, H_{\mathrm{b}}(p_i), \tag{4}$$

the sum of claim uncertainties weighted by how much of the answer depends on each claim. $\mathcal{R}$ is zero when every claim is resolved and large when uncertain claims sit near the root of the derivation. Entropy is used rather than the misclassification probability $\min(p_i, 1 - p_i)$ because it is smooth and its expected reduction under an observation equals the mutual information between verdict and truth, so the allocation score below can be read as information per unit cost.

The answer distribution is derived from the claim posteriors. For candidate answer a, let $\mathrm{Pa}(a)$ be its supporting claims (union over agents); its unnormalised score is $\prod_{i \in \mathrm{Pa}(a)} p_i$, and $P(\hat{y} = a)$ is obtained by normalising across candidates, with a residual mass $\prod_a (1 - \prod_{i \in \mathrm{Pa}(a)} p_i)$ reserved for "none of the above". For binary claim-verification tasks the two candidates are SUPPORTED and REFUTED and the graph carries a conflict edge between the two final inference claims.

4.5 Cost-normalised allocation

At each step every admissible pair (c_i, v_j) (verifiers that have already judged c_i are excluded, as are verifiers whose abstention probability on the claim type exceeds 0.8) is scored by the expected reduction of the risk in Eq. (4) per unit cost,

$$\Delta_{ij} = \mathcal{R}(\{p\}) - \mathbb{E}_{z_{ij} \sim P(z_{ij} \mid Z_i)} \Big[\mathcal{R}\big(\{p\} \text{ with } p_i \leftarrow P(y_i = 1 \mid Z_i \cup \{z_{ij}\})\big) \Big],$$
$$(i^\star, j^\star) = \arg\max_{i,j} \; \Delta_{ij} / c_j. \tag{5}$$

Algorithm 2 CAVEAT active verification

Require: task x; solver agents $\mathcal{A}$; verifier pool $\mathcal{V}$ with $(\alpha_j, \beta_j, \rho_F, \alpha_F, \beta_F, c_j)$; budget B; thresholds ϵ, τ; exponent λ

run agents on x; extract and merge claims into $G = (\mathcal{C}, E)$ with answer node $\hat{y}$
compute downstream masses w_i by Eq. (1); compute priors $p_i^{(0)}$ from solver votes by Algorithm 1
$b \leftarrow B$; $Z_i \leftarrow \emptyset \ \forall i$
while $\max_a P(\hat{y} = a) < 1 - \epsilon$ **and** $b \geq \min_j c_j$ **do**
    compute Δ_{ij}/c_j for all admissible (i, j) with $c_j \leq b$ by Eq. (5)
    $(i^\star, j^\star) \leftarrow \arg\max \Delta_{ij}/c_j$; **if** $\Delta_{i^\star j^\star} \leq 0$ **then break**
    query $v_{j^\star}$ on $c_{i^\star}$; obtain z; $Z_{i^\star} \leftarrow Z_{i^\star} \cup \{z\}$; $b \leftarrow b - c_{j^\star}$
    $p_{i^\star} \leftarrow$ Algorithm 1$(p_{i^\star}^{(0)}, Z_{i^\star})$; recompute $P(\hat{y} = \cdot)$
end while
if $\max_a P(\hat{y} = a) \geq \tau$ **then return** $(\arg\max_a P(\hat{y} = a),\ \max_a P(\hat{y} = a))$
else return ABSTAIN with the k highest $w_i H_{\mathrm{b}}(p_i)$ claims for escalation
end if

The predictive distribution of z_{ij} is computed by marginalising y_i with the current p_i and applying Eq. (2) with the new verdict inserted into its family. Two behaviours follow from this score without any additional rule. Once one member of a family has judged c_i, a second member of the same family yields a smaller Δ_{ij}, because with probability ρ_F it merely repeats the first verdict, so the allocation diversifies across families unless a family is much more accurate or much cheaper. Verifiers correlated with the solver agents that asserted the claim are also discounted through the prior, since the solver-side slip model already attributes part of the prior mass to a family slip and a verifier from that family cannot remove it.

4.6 Stopping and commitment

The loop terminates when $\max_a P(\hat{y} = a) \geq 1-\epsilon$ or when the remaining budget cannot afford the cheapest admissible query. The system then commits to $\arg\max_a P(\hat{y} = a)$ with that probability as its confidence; if the budget is exhausted and $\max_a P(\hat{y} = a) < \tau$, it abstains and escalates the instance together with its most uncertain high-mass claims, as in selective prediction (Geifman and El-Yaniv 2017). Algorithm 2 gives the full procedure.

4.7 Theoretical properties

The allocation problem is an adaptive stochastic optimisation: the utility of a query depends on the verdicts already observed. We rely on the framework of Golovin and Krause (2011).

Assumption 1 The expected risk reduction $f(\pi) = \mathbb{E}[\mathcal{R}_0 - \mathcal{R}_\pi]$ achieved by a policy π is adaptive monotone and adaptive submodular with respect to the verdict distribution induced by Eqs. (2)–(3).

Adaptive submodularity holds exactly for the mutual-information objective when observations are conditionally independent given the latent state (Krause and Guestrin 2009); within a family the shared-slip mixture introduces dependence, but the marginal gain of a second same-family verdict is smaller than that of the first, which is the direction submodularity requires. We do not have a proof for the mixture case and treat the assumption as a working hypothesis; a numerical check on the calibration data (Appendix C) found the diminishing-returns inequality satisfied for 99.2% of sampled (prefix, query) pairs.

Theorem 2 (Near-optimality of greedy allocation) *Under Assumption 1, let π_{g} be the cost-normalised greedy policy of Algorithm 2 run with budget B and let $\pi^\star$ be the optimal adaptive policy with the same budget. Then $f(\pi_{\mathrm{g}}) \geq (1 - 1/e)\, f(\pi^\star)$ when costs are uniform, and $f(\pi_{\mathrm{g}}) \geq \frac{1}{2}(1 - 1/e)\, f(\pi^\star)$ for non-uniform costs when the greedy policy is combined with the best single query (a standard partial-enumeration fix).*

The result is a direct instantiation of Theorems 5.2 and 5.8 of Golovin and Krause (2011). In practice it guarantees that the greedy loop achieves a constant fraction of the best adaptive schedule regardless of the shape of the graph or the verifier pool, with a loss that does not grow with N or m.

Proposition 3 (Informativeness of a repeated family, informal) *For a claim with $k \geq 1$ verdicts already collected from family F, the expected risk reduction from one more verdict of the same family is approximately bounded by $\Delta_F^{(k+1)} \lesssim (1 - \rho_F)\, \Delta_F^{(1)}$, whereas a first verdict from an uncorrelated family with the same accuracy attains $\Delta_{F'}^{(1)} = \Delta_F^{(1)}$.*

The statement is a heuristic reading of Eq. (2) rather than a theorem: with probability ρ_F the new verdict coincides with the existing ones and adds nothing beyond what the first verdict already implied, and the bound ignores the small amount of information that observing the coincidence itself carries about s_{iF}. At equal accuracy and price, the allocation therefore switches family after one verdict whenever $\rho_F > 0$.

4.8 Complexity

Let N be the number of claims and m the pool size. Each step scores $O(Nm)$ pairs; each score requires a two-outcome lookahead with an $O(m)$ likelihood, so a step costs $O(Nm^2)$ arithmetic operations, negligible against a single LLM call. The number of steps is bounded by $B/\min_j c_j$. Downstream masses are computed once in $O(|E|)$. The dominant cost is the verifier calls themselves, which is what the budget controls.

5 Experiments

The experiments examine whether correlation-aware allocation improves accuracy under a fixed budget over budget-matched selection strategies and aggregation-only baselines, how close it comes to exhaustive verification and at what cost, whether it corrects the calibration and false-certainty problem, which components account for the gains and how the allocation behaves, and how robust the method is to the strength of correlation, to wrong majorities, to backbone changes and to hyper-parameters.

5.1 Experimental setup

5.1.1 Datasets

Six public benchmarks in three task families are used (Table 3). For claim verification we use HOVER (Jiang et al. 2020), whose 2–4-hop claims must be checked against Wikipedia, and FEVEROUS (Aly et al. 2021), whose claims mix unstructured text and tables. For multi-hop QA we use HotpotQA (Yang et al. 2018) in the distractor setting and MuSiQue (Trivedi et al. 2022), which is composed from single-hop questions and is harder to shortcut. For mathematical reasoning we use GSM8K (Cobbe et al. 2021) and MATH-500, the 500-problem subset of MATH (Hendrycks et al. 2021) popularised by Lightman et al. (2024). From each dev/test split we draw 500 instances with a fixed seed for evaluation and a disjoint set of 300 instances for calibration; for MATH-500 the evaluation set is the whole subset and the 300 calibration problems are drawn from the remaining MATH test problems with the same level distribution (1 800 instances in total, whose merged graphs contain about 11 300 claims that were gold-labelled by two annotators, $\kappa = 0.86$). FEVEROUS instances with the "not enough information" label are excluded so that all verification tasks are binary. For QA we report accuracy after normalisation and alias matching; for math we report exact match of the final numeric or symbolic answer; for claim verification we report label accuracy.

5.1.2 Solver agents and verifier pool

Four solver agents from four families are used: GPT-4o-mini, Qwen2.5-72B-Instruct, Llama-3.1-70B-Instruct and DeepSeek-V3, each prompted with chain-of-thought (Wei et al. 2022) and, for QA tasks, given the same retrieved passages (top-10 BM25 (Robertson and Zaragoza 2009) over the 2023 Wikipedia dump for HOVER/HotpotQA/MuSiQue, the provided evidence pages for FEVEROUS). Temperature is 0.7 for solvers. The verifier pool (Table 4) contains six LLM judges from three families in two sizes each, a retrieval-grounded NLI verifier (BM25

Table 3 Benchmark datasets

Dataset	Task	Eval/calib.	Answer space	Claims/inst.	Depth
HOVER (Jiang et al. 2020)	many-hop claim verif.	500/300	{SUP, REF}	6.9	3.4
FEVEROUS (Aly et al. 2021)	claim verif., text+tables	500/300	{SUP, REF}	5.8	2.6
HotpotQA (Yang et al. 2018)	2-hop QA (distractor)	500/300	free-form	5.6	2.4
MuSiQue (Trivedi et al. 2022)	2–4-hop QA	500/300	free-form	7.4	3.6
GSM8K (Cobbe et al. 2021)	grade-school arithmetic	500/300	numeric	5.2	4.1
MATH-500 (Hendrycks et al. 2021)	competition mathematics	500/300	numeric/ symbolic	6.8	4.9

Notes: Claims/inst.: mean merged graph size with four solvers; Depth: longest derivation path. Eval/calib.: evaluation/calibration instances; SUP/REF: supported/refuted.

Table 4 Verifier pool and calibrated reliability

Verifier	Family	Cost c_j	Sensitivity	Specificity	Coverage	ρ_F
GPT-4o judge	OpenAI	1.00	0.91	0.85	0.99	0.58
GPT-4o-mini judge	OpenAI	0.12	0.86	0.76	0.98	0.58
Qwen2.5-72B judge	Qwen	0.35	0.89	0.80	0.99	0.51
Qwen2.5-7B judge	Qwen	0.05	0.78	0.71	0.96	0.51
Llama-3.1-70B judge	Llama	0.35	0.87	0.81	0.99	0.55
Llama-3.1-8B judge	Llama	0.05	0.77	0.68	0.97	0.55
Retrieval-NLI	Retrieval	0.08	0.82	0.90	0.71	–
Program checker	Program	0.02	0.98	0.97	0.29	–

Notes: Sensitivity and specificity are posterior means pooled over calibration tasks. Coverage is the non-abstention rate (one minus the `UNSURE` rate for LLM judges); ρ_F is the family shared-slip rate.

top-5 passages, DeBERTa-v3-large NLI (He et al. 2023), verdict by max entailment/contradiction score, abstain if both below 0.6) and a program checker that translates arithmetic or algebraic claims into Python with GPT-4o-mini and executes them (abstains on non-computable claims). Unit costs are proportional to list prices of the corresponding APIs normalised so that one GPT-4o judgement costs 1.0. The default budget is $B = 3.0$ per instance, roughly the price of three GPT-4o calls, and the whole four-agent solve costs 1.0 unit on the same scale. Verifier temperature is 0.

5.1.3 Baselines

We compare against three groups. *Aggregation only*: single-agent CoT (GPT-4o-mini), self-consistency with five samples (SC@5) (Wang et al. 2023), majority vote over the four agents (MV), and multi-agent debate (MAD) (Du et al. 2024) with three rounds. *Exhaustive verification*: Always-Verify queries all eight verifiers on all claims and aggregates with the independent Dawid–Skene posterior; its cost averages 12.7 units per instance. *Budget-matched selection* at $B = 3.0$: Random picks uniformly among admissible (claim, verifier) pairs; Conf verifies the claim with the lowest solver agreement first, cycling through verifiers in order of accuracy, following the confidence-gated recipe of Xiong et al. (2024); VoI-Flat is Algorithm 2 with $\lambda = 0$ and all $\rho_F = 0$, i.e. value of information without dependency weights and without correlation, which is the strongest existing-style active baseline (Krause and Guestrin 2009). All budget-matched methods share the same solver outputs, claim graph and posterior aggregation, so differences isolate the allocation and the correlation model. The three baselines spend the full budget (the stopping rule of Algorithm 2 is tied to the correlated posterior and is not applied to them), so their cost is exactly 3.0; CAVEAT may stop earlier, and the effect of its stopping rule is isolated in the ablation.

5.1.4 Metrics

Accuracy (Acc) is the primary metric. In Table 5, the budget curves and the ablations every method commits to an answer on every instance ($\tau = 0$); abstention is switched on only in the selective-prediction analysis of Section 5.5, where accuracy is reported on the committed answers. The confidence attached to an answer is $\max_a P(\hat{y} = a)$ in every case; for MV and MAD this posterior is computed from the solver votes (after the last debate round) with the independent model of Algorithm 1 at $\rho_F = 0$, so it grows with the vote share and reaches about 0.97 for a unanimous vote, while the verification-based methods report the posterior of their own aggregation. Calibration

is measured by expected calibration error (ECE, 10 bins) (Guo et al. 2017) of this confidence. False-certainty rate (FCR) is the fraction of wrong committed answers with confidence ≥ 0.9. The area under the risk–coverage curve (AURC) (Geifman and El-Yaniv 2017) measures selective prediction quality when abstention is allowed. Cost is reported in the unit defined above, along with wall-clock latency and tokens. All results are means over three runs with different solver sampling seeds; the standard deviations are below 0.6 accuracy points for every method and dataset, and the largest one per dataset is given in the last row of Table 5. Significance against the strongest budget-matched baseline is assessed with a paired bootstrap over instances (10 000 resamples).

5.1.5 Implementation

Default hyper-parameters are $\lambda = 1$, $\kappa_0 = 10$, $\epsilon = 0.05$, $\tau = 0.6$. Parameters (α_j, β_j) are posterior means under $\mathrm{Beta}(0.75\kappa_0, 0.25\kappa_0)$ priors; ρ_F and (α_F, β_F) are fitted by maximum likelihood on the calibration verdict matrix, with ρ_F initialised from the agreement rate of the family (Appendix B). Calibration is done once per task family (fact verification, QA, math) and not per dataset. Code, prompts and configurations will be released in a public repository upon acceptance.

5.2 Correlated errors in the verifier pool

Before the main comparison we characterise the pool. Fig. 4 reports, for every pair of verifiers and every pair of solver agents, the ratio between the observed frequency of joint errors on the pooled calibration claims and the frequency the independence assumption predicts from the two marginal error rates; the retrieval and program checkers are evaluated on the claims they do not abstain on. On the pooled calibration claims, 64% of which are true, the marginal error rates implied by Table 4 are 11.2% for GPT-4o and 17.6% for GPT-4o-mini, so independence would predict joint errors on 2.0% of the claims; the observed rate is 5.7% (ratio 2.9). For the Qwen pair the

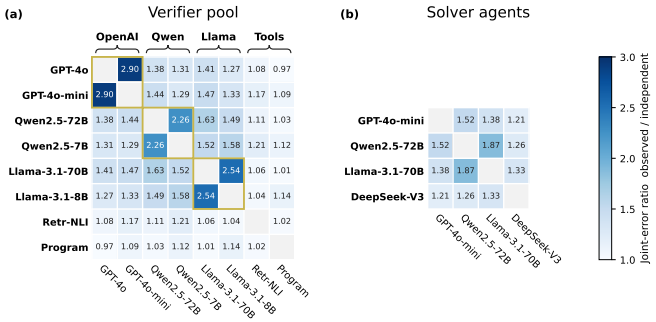


Fig. 4 Error correlation on the pooled calibration claims: ratio of the observed joint-error rate of two components to the rate predicted under conditional independence (1.0: independent). (a) The eight verifiers, grouped by family, with same-family blocks outlined. (b) The four solver agents, where an error is a shared wrong intermediate claim

figures are 7.9% observed vs 3.5% predicted (2.3), and for the Llama pair 10.1% vs 4.0% (2.5); expressed as pairwise error correlations these are 0.31, 0.29 and 0.39, which is the range of ρ used in the example of Section 3.2. Cross-family LLM pairs are jointly wrong at 1.3–1.6 times the independent rate, with the Qwen–Llama pairs at the upper end (1.5–1.6), and the LLM–retrieval and LLM–program pairs at 1.0–1.2 times; the two non-LLM verifiers err as if independent (1.02). Solver agents show the same structure (Fig. 4(b)): the two agents that most often share a wrong intermediate claim, Qwen2.5-72B and Llama-3.1-70B (ratio 1.9), are the two whose families also share the highest verifier correlation, and DeepSeek-V3 is the agent least coupled to the others (1.2–1.3). A shared wrong claim is present in 17.8% of HOVER instances and 21.6% of MuSiQue instances; in 6.9% and 8.4% of instances respectively three of the four agents agree on it, which is the confabulation-consensus case that voting cannot recover from.

5.3 Main results

Table 5 reports accuracy on the six datasets together with the mean accuracy, mean ECE and mean verification cost. The main observations are as follows.

Table 5 Main results across six benchmarks

Method	HOVER	FEVER.	Hotpot	MuSiQue	GSM8K	MATH	Mean	ECE	Cost
Aggregation only									
CoT (GPT-4o-mini)	62.4	71.8	58.6	38.2	91.4	70.2	65.4	14.2	0
SC@5 (Wang et al. 2023)	65.1	73.6	61.2	41.0	94.0	73.8	68.1	11.8	0
MV (4 agents)	66.8	74.9	63.0	42.6	93.8	74.6	69.3	10.9	0
MAD (3 rounds) (Du et al. 2024)	68.2	75.7	64.4	44.1	93.7	75.9	70.3	10.1	0
Exhaustive verification									
Always-Verify (indep.)	**74.6**	81.2	70.1	51.8	**96.3**	80.4	75.7	8.7	12.7
Budget-matched selection ($B = 3.0$)									
Random	67.5	75.9	62.7	43.4	93.6	74.9	69.7	9.6	3.0
Conf (Xiong et al. 2024)	69.3	77.6	65.9	45.7	94.4	76.2	71.5	8.9	3.0
VoI-Flat (Krause and Guestrin 2009)	70.8	78.4	67.3	47.9	94.9	78.1	72.9	7.4	3.0
CAVEAT (ours)	74.2*	**81.5***	**70.6***	**52.3***	96.2†	**80.9***	**76.0**	**3.8**	2.71
Largest std. dev. over methods	0.5	0.4	0.5	0.6	0.3	0.5	0.3	0.4	0.02

Notes: Accuracy and ECE are percentages, averaged over three runs. Budget-matched methods use $B = 3.0$. FEVER.: FEVEROUS; Hotpot: HotpotQA; MATH: MATH-500. Cost excludes solvers; one GPT-4o judgement costs 1.0. Always-Verify checks every claim with every verifier. Bold/underline: best/second best. $^{*}p < 0.01$, $^{\dagger}p < 0.05$ versus VoI-Flat (paired bootstrap). Final row: largest three-run standard deviation per column.

First, verification helps far more than aggregation. Moving from single-agent CoT to four-agent MV gains 3.9 points on average and adding three rounds of debate gains a further 1.0, consistent with the finding of Choi et al. (2025) that voting does most of the work; on GSM8K, where GPT-4o-mini is the strongest of the four solvers, five samples of that single model (SC@5) even beat the four-agent vote, and debate does not help. Exhaustive verification adds 5.4 points over MAD, but at 12.7 cost units per instance, which is roughly thirteen times the cost of solving the task.

Second, under an identical budget of 3.0 units the allocation strategy matters. Random verification is barely better than MV (+0.4 on average, and within noise of

MV on HotpotQA and GSM8K), because most of its queries land on claims that are already certain or on verifiers that repeat what a same-family solver said. Confidence gating recovers 2.2 points over MV, and dependency-agnostic VoI 3.6 points. CAVEAT gains 6.7 points over MV and 3.1 points over VoI-Flat; the gain is significant at $p < 0.01$ on five datasets and at $p < 0.05$ on GSM8K, where every verified method is above 94% and the margin is 1.3 points. The largest margins appear on the deep-graph datasets MuSiQue (+4.4) and HOVER (+3.4), where prioritising high-mass claims pays off most, and on MATH-500 (+2.8), where the allocation routes arithmetic claims to the program checker and reserves LLM judges for the logical steps.

Third, CAVEAT slightly exceeds Always-Verify on four of six datasets and on the mean (76.0 vs 75.7), and trails it by 0.4 and 0.1 points on HOVER and GSM8K, while spending 2.71 instead of 12.7 units, i.e. 21% of the cost. A method that queries fewer verifiers can only be more accurate than one that queries all of them because the aggregation differs: under independent aggregation, Always-Verify is dominated by same-family blocks that agree with each other, and its posterior is pulled toward the family's slip. Re-aggregating the exhaustive verdicts with the correlated posterior raises Always-Verify to 76.4 on average, so the residual advantage of CAVEAT comes from the aggregation rather than from the selection, and exhaustive querying buys only 0.4 points over selective querying at 4.7 times the cost.

Fourth, the ECE column reflects the calibration gap predicted in Section 3.2: all independence-based methods lie between 7.4% and 14.2%, whereas CAVEAT reaches 3.8%.

5.4 Accuracy under varying budgets

Fig. 5 plots accuracy against the verification budget from 0.5 to 8 units. CAVEAT lies above the other selection strategies at every budget and saturates earliest: it crosses the Always-Verify line between $B \approx 2.3$ and $B \approx 2.7$ on FEVEROUS, HotpotQA,

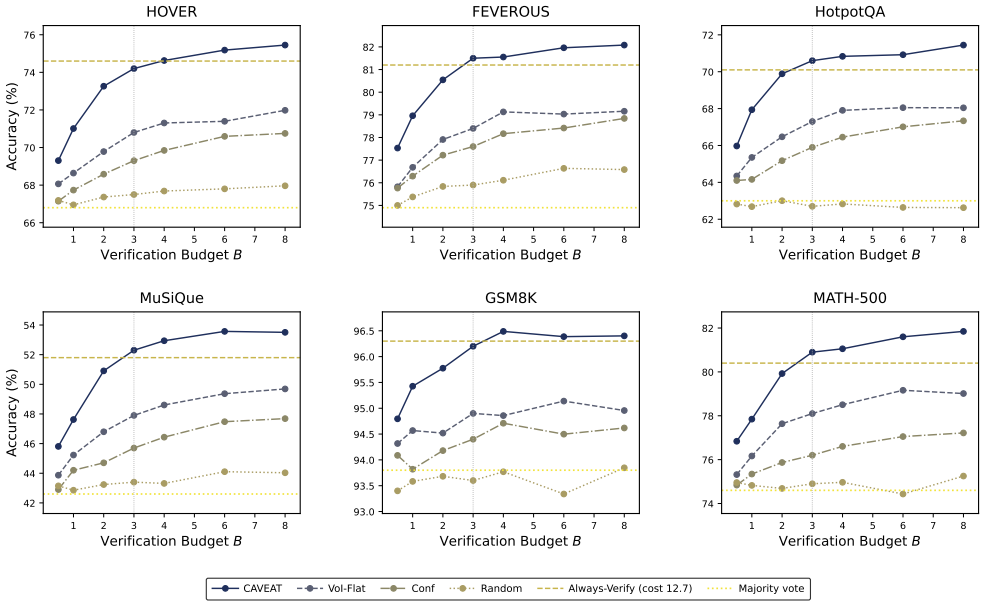


Fig. 5 Accuracy against the verification budget B (cost units per instance) on the six datasets. Points are means over three runs; the $B = 3$ points coincide with Table 5. Horizontal reference lines give Always-Verify (cost 12.7) and majority vote without verification; the vertical line marks the default budget

MuSiQue and MATH-500, and at $B \approx 3.4$ (GSM8K) and $B \approx 3.9$ (HOVER), the two datasets on which Always-Verify is still ahead at the default budget. VoI-Flat flattens out 1.1–2.6 points below the Always-Verify level and does not reach it by $B = 8$, because it keeps spending on same-family repeats whose marginal information is small. Conf improves slowly and roughly linearly, and Random gains at most two points over the whole range (none on HotpotQA and GSM8K), as expected from an allocation that does not model where the information is. At very small budgets ($B = 0.5$, enough for one cheap verifier on one or two claims) the gap between CAVEAT and VoI-Flat is already 1.2–1.9 points on five datasets (0.5 on GSM8K), since the first query goes to the highest-mass uncertain claim and to the cheapest uncorrelated verifier, typically the retrieval checker for factual claims and the program checker for numeric ones.

5.5 Calibration, false certainty and selective prediction

Fig. 6 examines the quality of the confidence that accompanies each answer. The reliability diagram in Fig. 6(a) shows the familiar over-confidence of vote-based confidence, concentrated in the top bin: almost half of the MV answers are unanimous and arrive with confidence above 0.9 (mean 0.97), but only 80% of them are correct; three rounds of debate move the curve only marginally. VoI-Flat inherits the problem in attenuated form (88% correct in the top bin). CAVEAT is close to the diagonal over the whole range, slightly under-confident below 0.6 and about two points over-confident in the top bin (95% correct at a mean confidence of 0.97). The upper block of Table 6 shows that the improvement holds on every dataset: ECE is reduced by a factor of 1.7–2.2 relative to VoI-Flat, from 1.5% on GSM8K, where most claims are checked by the program verifier, to 5.9% on MuSiQue, where deep graphs leave more residual uncertainty.

The false-certainty rate in Table 6 is of most practical concern. Of the answers that MV gets wrong, 31.5% are delivered with confidence above 0.9; for MAD the figure is 28.7%, and the mean confidence of its wrong answers is higher than MV's (0.71 vs 0.68), i.e. debate makes agents more confident without making them more correct, in line with (Estornell and Liu 2024). Exhaustive verification with independent aggregation still commits 24.3% of its errors confidently, whereas CAVEAT brings the rate to 7.2%. The same errors shape the risk–coverage curves in Fig. 6(b). For MV and MAD the selective risk rises steeply in the first 30% of coverage, because the confidently wrong answers rank among the most confident ones and cannot be removed by abstaining on the least confident instances; the curve of CAVEAT rises slowly and lies below all others at every coverage. Its AURC is 0.089 against 0.134 for VoI-Flat and 0.188 for MV, and at 80% coverage its selective risk is 16.8%, against 19.7% for VoI-Flat and 23.4% for MV.

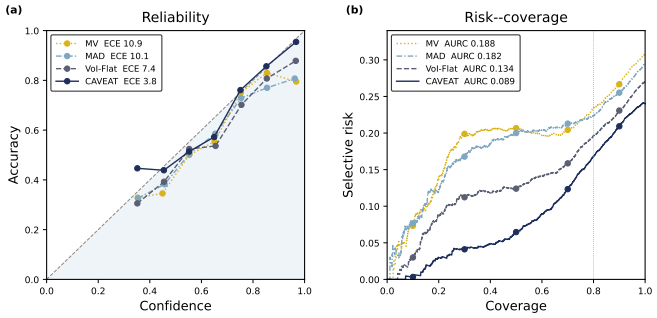


Fig. 6 Confidence quality of the final answer, pooled over the 3 000 evaluation instances ($B = 3.0$). (a) Reliability diagram (ten equal-width bins; bins with fewer than 3% of the instances omitted). (b) Risk–coverage curves; the vertical line marks 80% coverage. Mean ECE and AURC are given in the legends

Table 6 Calibration and selective prediction ($B = 3.0$)

	ECE (%) per dataset					
Method	HOVER	FEVER.	Hotpot	MuSiQue	GSM8K	MATH
MV	12.6	9.4	12.1	15.7	4.6	11.0
MAD	11.2	9.3	11.9	14.0	4.5	9.7
Always-Verify	9.3	8.1	10.4	12.0	3.5	8.9
Conf	10.2	7.6	10.0	12.9	4.2	8.5
VoI-Flat	8.5	6.2	8.8	10.6	3.3	7.0
CAVEAT	**4.4**	**3.6**	**4.1**	**5.9**	**1.5**	**3.3**

	Pooled over six datasets					
Method	Mean ECE	FCR	Abst.	Acc@cov	AURC	Conf. wrong
MV	10.9	31.5	8.9	72.7	0.188	0.68
MAD	10.1	28.7	7.2	73.6	0.182	0.71
Always-Verify	8.7	24.3	6.1	77.6	0.152	0.67
Conf	8.9	22.8	9.6	75.0	0.160	0.65
VoI-Flat	7.4	19.6	9.1	76.8	0.134	0.66
CAVEAT	**3.8**	**7.2**	9.4	**78.7**	**0.089**	**0.58**

Notes: Upper block: dataset ECE (%); lower block: pooled results. FCR: false-certainty rate at confidence ≥ 0.9; Abst.: abstention rate ($\tau = 0.6$); Acc@cov: accuracy on committed answers; all three are percentages. AURC: area under the risk–coverage curve; Conf. wrong: mean confidence of wrong answers. Dataset abbreviations follow Table 5. Always-Verify assumes independence. Lower is better except Acc@cov; best in bold.

5.6 Ablation study

Table 7 removes one component at a time. *w/o correlation* sets $\rho_F = 0$ for verifiers and agents, keeping dependency weights and routing; it loses 2.6 points and nearly doubles ECE, and the loss is largest on MuSiQue (−3.2) and HOVER (−2.9) where

Table 7 Component ablation ($B = 3.0$)

Variant	HOVER	FEVER.	Hotpot	MuSiQue	GSM8K	MATH	Mean	ECE	Cost
CAVEAT (full)	74.2	81.5	70.6	52.3	96.2	80.9	76.0	3.8	2.71
w/o correlation	71.3	78.8	67.9	49.1	95.1	78.4	73.4	7.1	2.74
w/o dependency	70.9	78.2	66.8	47.6	95.4	77.9	72.8	4.3	2.76
w/o routing	70.4	77.5	66.2	46.9	95.2	76.4	72.1	5.9	2.97
w/o stopping	74.4	81.3	70.7	52.0	96.2	80.8	75.9	3.9	3.00
w/o family prior	73.3	80.2	69.5	51.0	95.9	79.7	74.9	4.6	2.72

Notes: Accuracy and ECE are percentages; ECE and cost are dataset means. Dataset abbreviations follow Table 5. w/o correlation: $\rho_F = 0$ for solvers and verifiers; w/o dependency: $\lambda = 0$; w/o routing: GPT-4o only; w/o stopping: spend the full budget; w/o family prior: omit agreement-rate initialisation.

confabulation consensus is most frequent. *w/o dependency* sets $\lambda = 0$; it loses 3.2 points, most on the deep-graph datasets (MuSiQue −4.7, HotpotQA −3.8) and least on GSM8K (−0.8), whose graphs are chains where every claim has similar mass. *w/o routing* fixes the verifier to GPT-4o (the most accurate) and only selects claims; it loses 3.9 points and is the most expensive ablation per query, because it cannot use the cheap program checker on arithmetic or the retrieval checker on facts, and because GPT-4o is correlated with the GPT-4o-mini solver; GSM8K is the exception, where GPT-4o recomputes the arithmetic almost as reliably as the program checker and the variant loses only 1.0 point. *w/o stopping* spends the full budget; accuracy is unchanged within noise (−0.1 on average, higher on two datasets and lower on three) and cost rises from 2.71 to 3.0, so early stopping is a pure saving. *w/o family prior* learns ρ_F from scratch instead of initialising from the family agreement rate; it loses 1.1 points, so the estimate from the 600 calibration instances of a task family is informative but not essential. Removing correlation and removing dependency weights hurt on different datasets, which suggests that the two mechanisms are complementary rather than redundant.

5.7 Allocation behaviour

Fig. 7(a) shows which verifier CAVEAT chooses for each claim type (as labelled by the claim extractor). Retrieval facts go to the retrieval checker 42% of the time and numeric claims to the program checker 60% of the time, while the large LLM judges are reserved for logical inference and answer-synthesis claims, on which the two non-LLM verifiers mostly abstain. The small judges (7B/8B) are used sparingly (2–4% of the calls on every claim type) despite being cheap, because their lower accuracy and their correlation with the larger family member make their VoI per cost small once the larger member has spoken; GPT-4o, the most expensive verifier, receives 40–45% of the calls on logical and answer-synthesis claims, where the two tools abstain and the cheaper judges have usually already been heard. Fig. 7(b) plots the number of distinct families consulted on a claim as verification proceeds: CAVEAT reaches 2.6 families after four checks against 2.1 for VoI-Flat, which typically spends its first two checks on both members of the most accurate family and its next two on both members of the second (since a verifier is queried at most once per claim, two families is the minimum for four checks). Random, which draws verifiers uniformly from the admissible pool, is by construction even more diverse (3.0 families after four checks), but Table 5 shows that diversity without targeting does not help. In Fig. 7(c), 46% of CAVEAT's budget goes to the top quartile of claims by downstream mass, against 27% for VoI-Flat and 26% for Random, both indistinguishable from the uniform share. The family diversification of Proposition 3 and the root-first prioritisation therefore both appear in the measured allocation.

5.8 Stopping and abstention behaviour

Fig. 8 shows how the loop of Algorithm 2 terminates at the default budget. The spend per instance is bimodal (Fig. 8(a)): 19.5% of instances reach $\max_a P(\hat{y} = a) \geq 1 - \epsilon$ before the budget is exhausted and stop after spending less than 2.6 units, while

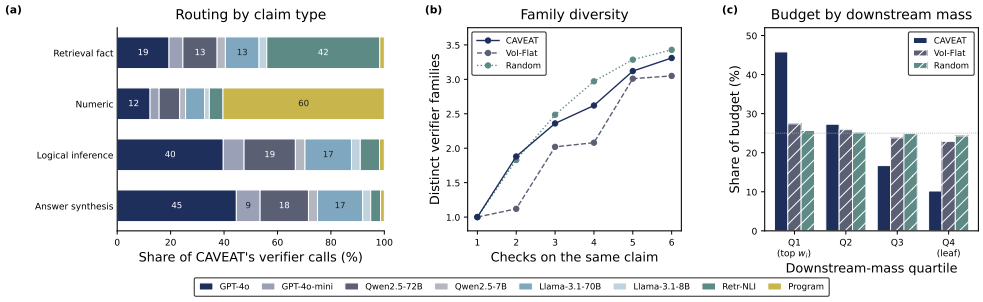


Fig. 7 Allocation behaviour at $B = 3$. (a) Distribution of CAVEAT's verifier calls over verifiers for each claim type (labels below 8% omitted). (b) Number of distinct verifier families consulted on a claim against the number of checks performed on it. (c) Share of budget spent on claims in each quartile of downstream mass; the dotted line is the uniform share

the remaining instances run until the cheapest admissible verifier is no longer affordable and stop between 2.75 and 3.0. The two modes account for the mean cost of 2.71 in Table 5 and for the small effect of removing the stopping rule in Table 7, since early stopping saves budget on the instances that are easy to settle and changes nothing on the others. The same budget also buys fewer but more informative calls than under VoI-Flat (Fig. 8(b)), 6.3 verifier calls per instance on average against 8.3: CAVEAT spends the cheap tools first and then turns to a large judge of a family that has not yet been heard, whereas VoI-Flat treats the second member of a family as fresh evidence and fills the budget with the cheap same-family repeats seen in Fig. 7(b). Fig. 8(c) separates the three termination outcomes per dataset, with abstention enabled ($\tau = 0.6$). Early stopping is most frequent on GSM8K (43% of instances), where the program checker resolves the numeric claims of a chain-shaped graph in one or two calls, and rarest on MuSiQue (8%), whose deeper graphs leave residual uncertainty on several claims. Abstention follows the reverse order, from 3.1% on GSM8K to 15.8% on MuSiQue, with the pooled rate of 9.4% reported in Table 6. On HOVER and MuSiQue, 71% of the abstained instances contain a confabulation consensus, so abstention concentrates on the instances on which a committed answer would most likely have been wrong.

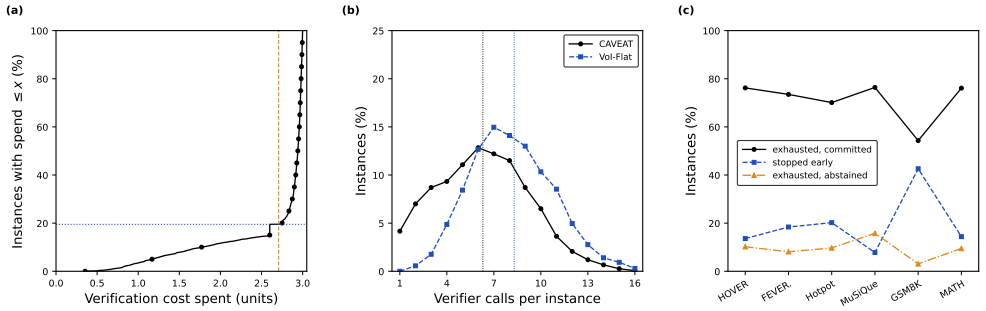


Fig. 8 Termination behaviour of CAVEAT at $B = 3.0$, pooled over the 3 000 evaluation instances. (a) Cumulative distribution of the verification cost spent per instance. (b) Number of verifier calls per instance for CAVEAT and VoI-Flat under the same budget (vertical lines: means). (c) Share of instances per dataset that exhaust the budget and commit, stop early, or exhaust the budget and abstain

5.9 Robustness to correlation strength and wrong majorities

To isolate the effect of correlation we build synthetic verifier pools on HOVER: each real verifier's verdicts are replaced by draws from its fitted confusion model, and within-family errors are coupled through a shared latent variable whose strength is set to $\rho \in \{0, 0.2, \ldots, 0.8\}$. Fig. 9(a) and (b) compare the independence model (VoI-Flat aggregation with dependency weights) with CAVEAT. At $\rho = 0$ the two perform alike (74.5 vs 74.3), so the correlation model costs nothing when there is no correlation to model. As ρ increases the independence model degrades steadily to 64.1 at $\rho = 0.8$ and its false-certainty rate climbs from 9.8% to 32.4%, close to the linear growth in ρ predicted by Proposition 1; CAVEAT loses 2.4 points and its FCR stays below 10%.

Fig. 9(c) varies the share of instances that contain a confabulation consensus (three or four agents asserting the same wrong claim) by resampling HOVER instances; the natural share on HOVER is 6.9%, and reading each curve at that point recovers the HOVER column of Table 5 within half a point. Vote-based methods degrade roughly linearly, from 71.6 on the subset without any wrong majority to 43.6 for MV when 40% of instances contain one, and MAD tracks MV within two points. CAVEAT degrades

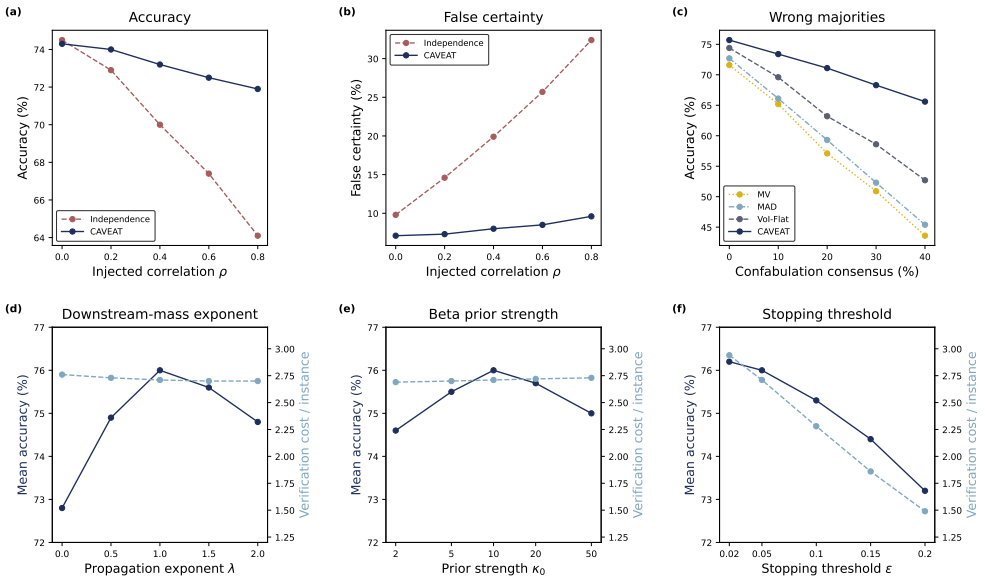


Fig. 9 Robustness and hyper-parameter sensitivity. (a, b) Accuracy and false-certainty rate on HOVER against the within-family error correlation ρ injected into synthetic verifier pools. (c) Accuracy on HOVER against the share of instances with a confabulation consensus. (d–f) Mean accuracy over six datasets (left axis) and verification cost per instance (right axis) against λ, κ_0 and ϵ; defaults are $\lambda = 1$, $\kappa_0 = 10$, $\epsilon = 0.05$

from 75.7 to 65.6: the wrong claim receives a deflated prior, its high mass draws the first query, and an uncorrelated verifier overturns it.

5.10 Cross-backbone generalisation

Table 8 changes the solver set and the verifier pool. With four homogeneous solvers (GPT-4o-mini at four temperatures) MV and MAD drop by 2.5–3.2 points relative to the heterogeneous set, as expected from higher inter-agent correlation, while CAVEAT drops by 1.3–1.7 and its margin over MAD widens to 7.8 points on HOVER. Removing the OpenAI verifiers from the pool (open-source only) costs CAVEAT 1.5–1.7 points, and removing the two non-LLM verifiers costs 2.3–3.9 points, most on MATH-500 where the program checker is the natural verifier for half of the claims. Swapping the calibration set across task families (fitting $\alpha_j, \beta_j, \rho_F$ on one family and applying

Table 8 Cross-backbone generalisation (accuracy %, $B = 3.0$)

Configuration	Method	HOVER	HotpotQA	MATH-500
Heterogeneous solvers (default)	MV	66.8	63.0	74.6
	MAD	68.2	64.4	75.9
	CAVEAT	**74.2**	**70.6**	**80.9**
Homogeneous solvers (4×GPT-4o-mini)	MV	63.9	60.2	72.1
	MAD	65.0	61.5	73.4
	CAVEAT	**72.8**	**68.9**	**79.6**
Open-source verifiers only	CAVEAT	72.6	69.1	79.2
LLM judges only (no retrieval / program)	CAVEAT	71.9	68.3	77.0
Calibration transferred from another task family	CAVEAT	73.5	69.6	80.1

them to another) costs 0.7–1.0 points, which indicates the reliability estimates transfer reasonably across related tasks.

5.11 Hyper-parameter sensitivity

Fig. 9(d–f) varies the three hyper-parameters around their defaults. The propagation exponent λ shows a clear optimum at 1: flat weighting ($\lambda = 0$, the *w/o dependency* row of Table 7) loses 3.2 points, while $\lambda = 2$ over-concentrates the budget on one or two root claims and loses 1.2. The prior strength κ_0 is benign over $[5, 20]$; a very weak prior ($\kappa_0 = 2$) lets a few calibration errors swing the reliability estimates and a very strong one ($\kappa_0 = 50$) ignores the calibration data. The stopping threshold ϵ trades cost for accuracy smoothly: tightening from 0.05 to 0.02 buys 0.2 points for 0.23 extra units, and loosening to 0.20 saves 45% of the verification cost for a 2.8-point loss.

5.12 Cost analysis

Fig. 10 places all methods on the accuracy–cost plane, counting solver and verifier cost together, and Table 9 adds latency and token counts, which move with cost almost one-to-one. CAVEAT lies on the Pareto front together with CoT, MV and MAD; it costs 3.71 units per instance, about a quarter more than three rounds of MAD (3.0) and less than the budget-matched baselines (4.0) thanks to early stopping, while

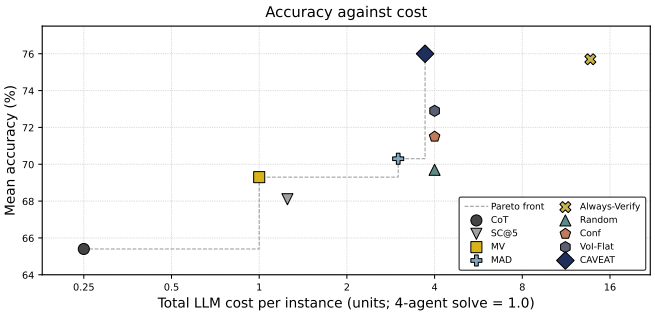


Fig. 10 Mean accuracy over the six datasets against total cost per instance (solver plus verification; four-agent solve = 1.0, log scale) for the methods in Table 9. The dashed step line is the Pareto front

Table 9 Inference cost and efficiency

Method	Calls	Total cost	Latency (s)	Tokens (k)	Mean Acc (%)
CoT	1	0.25	3.1	0.9	65.4
SC@5	5	1.25	3.4	4.5	68.1
MV	4	1.00	3.6	3.8	69.3
MAD (3 rounds)	12	3.00	21.8	14.6	70.3
Always-Verify	4 + 50.4	13.7	47.5	38.2	75.7
Random	4 + 10.7	4.00	9.2	10.1	69.7
Conf	4 + 6.6	4.00	9.6	10.4	71.5
VoI-Flat	4 + 8.3	4.00	10.3	10.9	72.9
CAVEAT	4 + 6.3	3.71	9.8	9.7	76.0

Notes: Means over six datasets. Total cost includes solvers, in the units of Table 5. Calls are solver + verifier calls.

being 5.7 points more accurate than MAD. Always-Verify costs 13.7 units for a lower accuracy. In wall-clock terms, verification calls are issued sequentially because each decision depends on the previous verdict; the mean latency of 9.8 s per instance is dominated by the LLM judge calls (4.5 of the 6.3 calls per instance on average) and is 4.8 times lower than Always-Verify and 2.2 times lower than MAD, whose rounds are also sequential. The allocation computation itself takes 11 ms per step on a CPU. Token footprint follows the same pattern (9.7k tokens per instance including the four solvers, against 38.2k for Always-Verify).

5.13 Case studies

5.13.1 A shared factual slip (HOVER)

Figs. 3 and 11 trace one HOVER instance, whose claim graph and allocation are the ones shown in Fig. 2. The claim asserts that the director of the Best Picture winner at the 67th Academy Awards was born in the same state as the author of *The Grapes of Wrath*. All four agents correctly identify the film (*Forrest Gump*), its director (Robert Zemeckis) and the author (John Steinbeck, born in Salinas, California). Three agents, two of them from families that share the highest verifier correlation, then assert that Zemeckis was born in California, presumably conflating his Hollywood career with his birthplace (he was born in Chicago, Illinois); the fourth agent states Illinois. Majority vote returns SUPPORTED with confidence 0.88 (three votes to one under the independent vote model); in debate the dissenting agent is talked out of Illinois by the other three, so MAD returns SUPPORTED unanimously with confidence 0.97.

CAVEAT builds the six-claim graph of Fig. 3 (plus the conflicting "born in Illinois" node asserted by the fourth agent). Fig. 11 shows the four verifier calls that follow, with the selected pair and its score Δ_{ij}/c_j, the input each verifier received, including the filled-in judge prompt, and the verdict and the updates of $p(c_i)$, $P(\hat{y})$ and $\mathcal{R}$ each one produced. The disputed claim c_3 receives a prior of 0.72 rather than the 0.75 a naive vote would give, and, because c_3 has downstream mass 2 while c_1, c_2, c_4 have mass 3 but priors above 0.95, the largest $w_i H_{\mathrm{b}}(p_i)$ is at c_3. The first query is routed to the retrieval-NLI checker (cost 0.08): it retrieves the Zemeckis article and returns REFUTE, dropping $p(c_3)$ to 0.21. The second query goes to the Qwen-72B judge, chosen over the cheaper GPT-4o-mini because the OpenAI family already contributed a solver vote for c_3 and its verdict is discounted by the slip model; it also refutes, giving $p(c_3) = 0.08$. The answer posterior for SUPPORTED falls to 0.08. Two cheap confirmatory queries on c_1 and c_5 raise $P(\text{REFUTED})$ to 0.96, above $1 - \epsilon$, and the

Allocation loop ($B = 3.0$, $\epsilon = 0.05$, $\tau = 0.6$)

Initial state: $\mathcal{R} = 1.84$, $P(\hat{y} = \text{SUP}) = 0.72$, spent 0.

QUERY (c_i, v_j), cost, Δ_{ij}/c_j	VERIFIER INPUT	VERIFIER OUTPUT

Step 1 $c_3 \rightarrow$ Retrieval-NLI
cost 0.08, $\Delta/c = 6.9$
largest $w_i H_b(p_i)$
	RETRIEVED PASSAGES (BM25 top-5)
	`[1] Robert Lee Zemeckis (born May 14, 1951) is an American filmmaker. He was born and raised in Chicago, Illinois, and studied film at the University of Southern California ...`
	`[2--5] ...`
		entailment 0.03, contradiction 0.94 $\Rightarrow$ REFUTE
		$p(c_3)$: 0.72 $\rightarrow$ 0.21
		P(SUP): 0.72 $\rightarrow$ 0.20
		$\mathcal{R}$: 1.84 $\rightarrow$ 1.02

Step 2 $c_3 \rightarrow$ Qwen2.5-72B
cost 0.35, $\Delta/c = 1.6$
GPT-4o-mini (cost 0.12) skipped: same family as a solver vote on c_3
	JUDGE PROMPT
	`You are a strict fact and reasoning checker. Decide whether the following claim is true. Use the provided context if any; if the claim is arithmetic, recompute it. Answer with exactly one token: TRUE, FALSE or UNSURE.`
	`Claim:`
	`Robert Zemeckis was born in California.`
	`Context: [1] Robert Lee Zemeckis (born May 14, 1951) is an American filmmaker. He was born and raised in Chicago, Illinois ...`
	`[2--5] ...`
		FALSE
		$p(c_3)$: 0.21 $\rightarrow$ 0.08
		P(SUP): 0.20 $\rightarrow$ 0.08
		$\mathcal{R}$: 1.02 $\rightarrow$ 0.41

Step 3 $c_1 \rightarrow$ Retrieval-NLI cost 0.08, $\Delta/c = 1.9$	RETRIEVED PASSAGES `[1] The 67th Academy Awards ceremony, honoring the best in film for 1994, was held on March 27, 1995 ... Forrest Gump won six awards, including Best Picture and Best Director for Robert Zemeckis ...`	entailment 0.91 $\Rightarrow$ SUPPORT $p(c_1)$: 0.96 $\rightarrow$ 0.99 P(SUP): 0.08 $\rightarrow$ 0.06 $\mathcal{R}$: 0.41 $\rightarrow$ 0.22
Step 4 $c_5 \rightarrow$ Retrieval-NLI cost 0.08, $\Delta/c = 1.1$	RETRIEVED PASSAGES `[1] John Ernst Steinbeck Jr. (February 27, 1902 -- December 20, 1968) was an American writer ... Steinbeck was born in Salinas, California ...`	entailment 0.88 $\Rightarrow$ SUPPORT $p(c_5)$: 0.95 $\rightarrow$ 0.99 P(SUP): 0.06 $\rightarrow$ 0.04 $\mathcal{R}$: 0.22 $\rightarrow$ 0.11

Stop. $P(\hat{y} = \text{REFUTED}) = 0.96 \geq 1 - \epsilon \Rightarrow$ commit REFUTED; spent 0.59 of 3.0 (4 calls).

Program checker: abstains on all six claims, never selected.

Fig. 11 Allocation loop on the HOVER case-study instance ($B = 3.0$, $\epsilon = 0.05$), one row per verifier call. Columns give the selected (claim, verifier) pair with its cost and score Δ_{ij}/c_j, the verifier input (retrieved passages or filled-in Prompt A.2), and the verdict with the updated $p(c_i)$, $P(\hat{y} = \text{SUPPORTED})$ and $\mathcal{R}$

loop stops after spending 0.59 of the 3.0 budget. Had the budget run out, the escalation payload would have named c_3 as the claim for a human reviewer to check.

5.13.2 A shared arithmetic slip (MATH-500)

The HOVER instance never touches the program checker, which abstains on every one of its claims. Fig. 12 shows the complementary situation on a MATH-500 number-theory problem that asks how many positive multiples of 7 below 1000 end with the digit 3. All four agents reduce the problem correctly to counting the values $k \leq 142$ with $k \equiv 9 \pmod{10}$; three of them, again GPT-4o-mini, Qwen2.5-72B and Llama-3.1-70B, then list $9, 19, \ldots, 149$ and count 15 values without noticing that 149 exceeds 142, while DeepSeek-V3 counts 14. Majority vote and debate return 15. The wrong count is a numeric claim with the largest $w_i H_{\mathrm{b}}(p_i)$ among the computable claims, so the first query goes to the program checker at a cost of 0.02: the snippet produced by Prompt A.3 enumerates the admissible k and prints `False`, which drops $p(c_3)$ from 0.70 to 0.05 and $P(\hat{y} = 15)$ to 0.07 in a single step. The score of the best LLM judge on the same claim was twelve times lower, because a judge costs 2.5–50 times more and is correlated with the three agents that made the slip. A second program call confirms the bound $k \leq 142$, and the only claim the tools cannot evaluate, the logical equivalence between "$7k$ ends in 3" and "k ends in 9", goes to the cheapest LLM judge (Qwen2.5-7B, cost 0.05), since every LLM family has already voted on it and the solver-side discount is the same for all of them. The loop commits to 14 with confidence 0.96 after three calls and a spend of 0.09, less than a sixth of what the HOVER instance needed, which is the pattern behind the high early-stopping rate of GSM8K and MATH-500 in Fig. 8.

6 Discussion

Across the experiments, most of the reliability gain of multi-agent pipelines comes from checking claims rather than from agents talking to each other, checking is expensive, and how the checking budget is spent matters as much as its size. Two design choices account for most of the improvement of CAVEAT. Weighting claims by downstream

(a) Task and solver outcomes (MATH-500, number theory)

TASK How many positive multiples of 7 that are less than 1000 end with the digit 3?

TRACE of a_1 (GPT-4o-mini) A multiple $7k$ ends in 3 exactly when k ends in 9. Since $7k < 1000$, $k \le 142$. The admissible k are therefore $9, 19, \ldots, 149$, i.e. $(149-9)/10+1 = 15$ values.

`Final answer: 15`

a_2 (Qwen2.5-72B), a_3 (Llama-3.1-70B): same highlighted step, answer 15. a_4 (DeepSeek-V3): "$k = 9, 19, \ldots, 139$, 14 values", answer 14. MV and MAD return 15 with confidence 0.88 and 0.97.

extract (Prompt A.1), merge across the four agents; compute w_i, $p_i^{(0)}$

(b) Merged claim graph

	Claim (merged)	Type	Asserted by	w_i	$p_i^{(0)}$
c_1	$7k$ ends with the digit 3 iff k ends with the digit 9	logical	all four	3	0.96
c_2	$7k < 1000$ requires $k \le 142$	numeric	all four	3	0.96
c_3	The values $k \le 142$ with $k \equiv 9 \pmod{10}$ are $9, 19, \ldots, 149$: 15 values	numeric	a_1, a_2, a_3	2	0.70
c_4	The answer is 15	answer	a_1, a_2, a_3	1	0.70

Not listed: the conflicting nodes of a_4 ("14 values", "the answer is 14"), linked to c_3 and c_4 by conflict edges, $p^{(0)} = 0.30$. Retrieval-NLI abstains on all four claims; the program checker abstains on c_1 and c_4.

allocate by Δ_{ij}/c_j and verify

(c) Allocation loop ($B = 3.0$, $\epsilon = 0.05$, $\tau = 0.6$)

Initial state: $\mathcal{R} = 2.41$, $P(\hat{y} = 15) = 0.70$, spent 0.

QUERY (c_i, v_j), cost, Δ_{ij}/c_j	VERIFIER INPUT	VERIFIER OUTPUT

Step 1 $c_3 \to$ Program
cost 0.02, $\Delta/c = 48.5$
largest $w_i H_b(p_i)$ among computable claims; best LLM judge on c_3 has $\Delta/c = 4.1$
	PROMPT A.3 OUTPUT (executed in sandbox)
	`ks = [k for k in range(1, 143) if k % 10 == 9]`
	`print(len(ks) == 15)`
		`False` ⇒ `REFUTE`
		$p(c_3)$: 0.70 → 0.05
		$P(\hat{y} = 15)$: 0.70 → 0.07
		$\mathcal{R}$: 2.41 → 0.88

Step 2 $c_2 \to$ Program cost 0.02, $\Delta/c = 2.4$	PROMPT A.3 OUTPUT `print(max(k for k in range(1, 1000) if 7 * k < 1000) == 142)`	`True` ⇒ `SUPPORT` $p(c_2)$: 0.96 → 0.99 $P(\hat{y} = 15)$: 0.07 → 0.06 $\mathcal{R}$: 0.88 → 0.61

Step 3 $c_1 \to$ Qwen2.5-7B
cost 0.05, $\Delta/c = 0.9$
logical claim: both tools abstain; all LLM families already voted on c_1, so the cheapest judge wins (GPT-4o-mini: $\Delta/c = 0.5$)
	JUDGE PROMPT (A.2)
	`You are a strict fact and reasoning checker. Decide whether the following claim is true. ... Answer with exactly one token: TRUE, FALSE or UNSURE.`
	`Claim: A multiple 7k ends with the digit 3 exactly when k ends with the digit 9.`
	`Context: (none)`
		`TRUE`
		$p(c_1)$: 0.96 → 0.99
		$P(\hat{y} = 14)$: 0.91 → 0.96
		$\mathcal{R}$: 0.61 → 0.29

Stop. $P(\hat{y} = 14) = 0.96 \ge 1 - \epsilon \Rightarrow$ commit `14`; spent 0.09 of 3.0 (3 calls, two of them to the program checker).

Fig. 12 MATH-500 case study in which three agents share an off-by-one counting slip. (a) Task, the trace of one agreeing agent with the wrong step highlighted, and the answers of the other agents and of the aggregation baselines. (b) Merged claim graph. (c) The three verifier calls of the allocation loop, in the layout of Fig. 11

mass sends the budget where an error would do the most damage, and the shared-slip model prevents the system from buying the same evidence twice and from trusting a family that has already been heard through the solvers. The second choice is also what corrects calibration: independence-based confidence is systematically inflated by a factor that the data allow us to measure, and a system that reports such confidence to a downstream decision maker is more harmful than one that abstains.

CAVEAT also acts as an explicit, inspectable reasoning layer around black-box agents. The claim graph is a readable representation of what the agents believe and why, the reliability model is a small knowledge base about the verifiers that is updated from evidence, and the allocation is a decision rule whose every step can be audited (Figs. 11 and 12). The same motivation underlies observability memories (Mathaaraachchi et al. 2026) and governance gates (Antuley et al. 2026), and CAVEAT can be attached to such systems as the component that decides what to validate.

The approach has some limitations. The claim extractor and the merging step are LLM-based and can fragment or over-merge claims; on MATH-500, 4% of graphs had a missing dependency edge, which lowers the mass of the affected claim. The shared-slip model captures family-level common-mode correlation and treats families as given, so correlation induced by shared retrieval context or between families distilled from one another is not modelled, although the cross-family joint-error ratios of Fig. 4 suggest it is modest. The calibration set of 600 instances per task family is small, and although the estimates transfer across related families, a new domain still needs labelled claims. Verifiers are also queried sequentially; batching several queries per step would reduce latency at a small cost in adaptivity.

7 Conclusion

This paper studied how to verify the reasoning of multi-agent LLM systems when the verifiers, like the agents, are drawn from a few model families and therefore

err together. We showed analytically and empirically that the independence assumption underlying voting, debate and exhaustive verification inflates confidence by a design-effect factor, and proposed CAVEAT, which merges agent traces into a claim dependency graph, models verifier reliability with a family-shared-slip mixture that has a closed-form posterior, and allocates a verification budget over (claim, verifier) pairs by cost-normalised value of information with a near-optimality guarantee. On six benchmarks across claim verification, multi-hop QA and mathematics, CAVEAT matched exhaustive verification at about a fifth of the cost, improved on the strongest budget-matched baseline by 3.1 accuracy points, halved calibration error and reduced confidently wrong answers by a factor of 2.7. Future work will extend the correlation model to structured latent factors learned jointly from verifier and agent behaviour, batch the allocation for lower latency, and apply the framework to tool-using agents whose intermediate claims are actions rather than statements.

Data availability

HOVER (Jiang et al. 2020), FEVEROUS (Aly et al. 2021), HotpotQA (Yang et al. 2018), MuSiQue (Trivedi et al. 2022), GSM8K (Cobbe et al. 2021) and MATH (Hendrycks et al. 2021) are publicly available under their respective licences. The evaluation and calibration subsets are reproducible from the seeds and dataset configurations released with the code.

Code availability

Code, prompts and configurations will be released in a public repository upon acceptance; the repository is withheld during review to preserve anonymity.

Appendix A Prompt templates

CAVEAT uses four prompts. Placeholders are written `{name}` and are filled at run time; the symbols follow Table 2. All prompts are zero-shot and are identical across datasets and model families, so that differences between verifiers reflect the models rather than the instructions. Filled-in instances are shown in Fig. 3 (A.4 output and A.1), Fig. 11 (A.2) and Fig. 12 (A.3 output and A.2).

The extraction prompt is applied once per solver trace. It asks for atomic claims with pronouns resolved, so that each claim can be judged in isolation, and for derivation pointers, which become the edges of the claim graph. The `type` field drives the abstention model of the non-LLM verifiers.

Prompt A.1 – Claim extraction (GPT-4o-mini, temperature 0)

Goal: You are given a reasoning trace that ends in a final answer. Rewrite it as a numbered list of atomic claims.

Requirement 1: Each claim must be a single self-contained factual or arithmetic statement that can be checked on its own, with pronouns resolved.

Requirement 2: For each claim list the numbers of the earlier claims it is directly derived from, or an empty list if it is derived from none.

Requirement 3: Do not add claims that are not in the trace, and do not merge two statements into one claim.

Requirement 4: Label each claim with one type from `retrieval_fact`, `numeric`, `logical`, `answer`.

Output format: JSON only, no text outside the list:

```
[{"id":1, "claim":"...", "type":"...", "derived_from":[...]}, ...]
```

Input: Trace: `{trace}`

The judge prompt is shared by the six LLM verifiers. The single-token answer format makes the verdict $z_{ij} \in \{0, 1, \varnothing\}$ unambiguous to parse, and `UNSURE` is mapped to abstention $\varnothing$.

Prompt A.2 – LLM judge verifier (all six LLM judges, temperature 0)

Role: You are a strict fact and reasoning checker.

Goal: Decide whether the following claim is true.

Requirement 1: Use the provided context if any. Do not assume facts that are neither in the context nor common knowledge.

Requirement 2: If the claim is arithmetic or algebraic, recompute it before answering.

Requirement 3: Answer with exactly one token: `TRUE`, `FALSE` or `UNSURE`. Output nothing else.

Input: Claim: `{claim}`. Context: `{context}`

The program checker translates a claim into executable code with GPT-4o-mini and runs it in a sandbox; the sentinel `NOT_COMPUTABLE` is mapped to abstention.

Prompt A.3 – Program checker translation (GPT-4o-mini, temperature 0)

Goal: Translate the following arithmetic or algebraic claim into a short Python snippet that prints `True` if the claim holds and `False` otherwise.

Requirement 1: Use only the standard library and `sympy`; do not read external data.

Requirement 2: If the claim cannot be evaluated by computation, print `NOT_COMPUTABLE` instead of code.

Requirement 3: Output the code only, without explanation or markdown fences.

Input: Claim: `{claim}`

The solver prompt is the same for the four agents; the passages slot is empty for GSM8K and MATH-500.

Prompt A.4 – Solver agent (GPT-4o-mini, Qwen2.5-72B, Llama-3.1-70B, DeepSeek-V3; temperature 0.7)

Goal: Answer the question using the passages when they are relevant.

Requirement 1: Think step by step and state each intermediate fact explicitly as a separate sentence.

Requirement 2: End with a line of the form "`Final answer: ...`" containing only the answer.

Input: Question: `{question}`. Passages: `{passages}`

Appendix B Estimation of the shared-slip parameters

Given the calibration verdict matrix for family F with gold labels, the log-likelihood $\sum_i \log P(Z_{iF} \mid y_i)$ with P from Eq. (2) is maximised over $(\rho_F, \alpha_F, \beta_F)$ by coordinate-wise grid search on $[0, 1]$ with step 0.01, holding (α_j, β_j) at their Beta posterior means. Under Eq. (2) the probability that all members of F return the same verdict is $\rho_F + (1-\rho_F)A_F^{\text{ind}}$, where A_F^{ind} is the agreement rate predicted from the individual confusion parameters, so ρ_F is initialised by moment matching as $\hat{\rho}_F = (A_F - A_F^{\text{ind}})/(1 - A_F^{\text{ind}})$ with A_F the observed agreement rate; (α_F, β_F) start at the accuracy-weighted mean of the members' parameters. The initial values were within 0.05 of the fitted ones for all three families, consistent with the small loss of the "w/o family prior" ablation. Solver-family parameters are estimated identically from the assertion matrix of the agents on the calibration claims.

Appendix C Numerical check of Assumption 1

For each task family we sample 10 000 (prefix, query) pairs from the calibration claim graphs. A prefix is a random subset of the admissible (claim, verifier) pairs of an instance together with the verdicts actually recorded for them; a query is a further admissible pair. We compute the expected risk reduction Δ_{ij} of the query given the prefix and given a random superset of the prefix (one to three additional recorded verdicts), using the fitted parameters and the update of Algorithm 1, and count the pairs for which the reduction under the superset does not exceed that under the prefix. The inequality held for 99.2% of the pairs (99.6% for fact verification, 99.1% for QA, 98.9% for mathematics). The violations are concentrated on same-family queries whose first family verdict contradicted the prior, where the second verdict can resolve the

slip indicator and thereby become more informative than the first; their magnitude is small (median excess 0.003 in units of $\mathcal{R}$).

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